

AI-POWERED CAD ANALYSIS REPORT

Generated: 12 Aug 2026

Geometry to Process to Cost · Fully AI-reasoned

53

BUMPER

1691 × 647 × 528 mm · 2059.9 cm³ · AI 5.562 kg / Steel 16.170 kg

Manufacturability: 53/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	1690.9 × 647.4 × 528.1 mm	Surface Area	16261.7 cm²
Volume	2059.90 cm³	Weight (AI)	5.562 kg
Weight (Steel)	16.170 kg	Weight (Plastic)	2.163 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
free_form_surface	492	High	492 B-spline faces — large organic cosmetic Class-A shell (bumper fascia)
thin_shell	1	High	Uniform ~2.5 mm wall shell, fill ratio 0.0036 !' hollow moulded
undercut	4	High	4 undercut faces + 2 zero-draft faces on [0,0,1] draw !' side a
threaded_feature	1	Medium	Threaded feature detected (likely moulded boss/insert for fixings)
planar_face	6	Low	6 planar faces (mounting/reference surfaces)

§3 — Material Analysis

AI-suggested

Primary Material: PP Impact Copolymer (PP-B) (85% confidence)

Large 1.7 m automotive bumper fascia with a uniform ~2.5 mm thin shell — the industry-standard resin for exterior bumper fascias is impact-modified polypropylene (PP-B / TPO). It offers low cost, good impact resilience, paintability with primer, and easy flow across the very large flow length. PC/ABS is the alternative for higher-gloss/stiffer parts but PP dominates fascia moulding.

Alternatives: PC/ABS Blend (automotive grade) (40%) · ABS (20%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Injection Moulding	injection_moulding	98%	0.0167	User-forced and consistent with geometry: hollow thin-wall (2.5 mm) large organic Class-A shell at 200,000/yr ! high-volume injection moulding fascia.

§5 — Manufacturability Risks (Score: 53/100)

Severity	Feature / Area	Description	Recommended Action
High	Undercuts (4) + zero-draft (2)	Undercut faces on the [0,0,1] draw direction cannot release without side actions or lifters, and 2 zero-draft faces risk drag/scuffing on ejection.	Add hydraulic slides/lifters for the undercut regions (mounting tabs, lamp apertures) and apply "e1.5° draft to the zero-draft wa
High	Flow length vs wall	1.69 m flow length at 2.5 mm wall gives a high L/t ratio ! short-shot and high injection pressure especially at flow ends.	Use sequential valve-gate hot runner (3–5 m-apertures) skimp wall "e2.5 mm and consider for flow; balance fill with Moldflow.
Medium	Warpage on large shell	Large thin cosmetic shell with non-uniform wall (2.53–3.54 mm, Å high) is prone affecting fit to body.	Even out wall transitions, add hidden ribs for stiffness ("d60% of wall), balanced cooling circuits, and controlled pack profile.
Medium	Sink & weld lines	Thick sections at bosses/thread features and multi-gate fill create sink marks and weld lines on the Class-A surface.	Core out thick bosses, gate to push weld lines to non-visible zones, and use grained finish/paint to mask cosmetic defects.
Low	Threaded feature	Moulded thread/insert region requires unscrewing core or brass insert.	Prefer heat-staked/self-tapping boss instead of moulded thread where possible.

§6 — DFM Issues (injection_moulding)

Severity	Area	Description	Impact	Fix
High	Undercuts / zero-draft	4 undercut faces and 2 zero-draft faces on the Z draw cannot demould with a simple two-plate tool.	Requires multiple hydraulic slides/lifters ! +£40–80 and longer cycle.	Add side actions for undercuts and "e1.5° draft to all vertical Class-A walls.
High	Flow length / wall	1690 mm flow length at 2.5 mm wall ! very high L/t, short on stage press and pressure risk.	Short shots, high scrap, need large tonnage press and multi-gate hot runner.	Sequential valve-gate hot runner, PP-B high-flow grade, verify with Moldflow.
Medium	Wall uniformity	Wall varies 2.53–3.54 mm — non-uniform sections drive differential cooling.	Warp age / waviness ! poor body-gap fit, cosmetic rejects.	Equalise wall thickness, gentle transitions, balanced conformal cooling.
Medium	Class-A cosmetics	Weld lines and sink at gates/bosses on a visible painted surface.	Visible defects after paint rework/rejects.	Position gates to hide weld lines, core out bosses, tune pack pressure.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC

£6.50

MOST LIKELY

£9.50

CONSERVATIVE

£15.00

Net Weight	1.854 kg	Material	mat-pp
Recommended Process	injection_moulding	Cycle Time	0.0167 hr/part
Setup Time	0.750 hr	Operations	1 ops
Operations Detail	Injection moulding (large fascia, single cavity) (mach-haas-vf2, 0.0167 hr)		

Process-Specific Parameters

Casting Subtype	hpdc
Die/Mould Cost	£300,000
Die Life	100,000 shots
Cavities	1
Yield	90.0%
Flash Weight	0.000 kg
Yield	85.0%
Die Cost	£180,000
Strokes	7
Cavities	1
Wall Thickness	2.5 mm
Mould Cost	£312,157
Mould Life	1,000,000 shots
Projected Area	10946.8 cm²

§8 — AI Analysis Explanation

The part is a large automotive bumper fascia: a 1.69 m organic Class-A shell (492/498 B-spline faces) with a thin, near-uniform ~2.5 mm wall and a fill ratio of 0.0036 confirming a hollow moulded shell — this is classic large-part injection moulding. Resin is undecided from geometry; I selected PP Impact Copolymer (PP-B) as the industry-standard bumper fascia material (impact-tough, low cost, paintable), with PC/ABS as the higher-stiffness alternative. Net part weight in PP is 2.163 kg. Projected area (10,946.8 cm²) and wall (2.5 mm) are used verbatim per the deterministic rules. At 200,000/yr a single large-cavity dedicated tool is appropriate; mould cost ~£200k. Main risks are the 4 undercuts/2 zero-draft faces needing side actions, the very high flow-length-to-wall ratio, and warpage/weld-line cosmetics on the painted surface. Manufacturability score fixed at 53.

§9 — Analysis Limitations & Assumptions

1. Resin (materialId), cavity pressure, cool/fill/pack times and runner weight are UNDECIDED from geometry — confirm the moulding resin to finalise cycle time and machine tonnage.
2. Cycle time estimated from thin-wall large-part heuristics, not a full Fourier cooling + machine-tonnage calc (needs resin thermal properties).
3. Machine tonnage: projected area 10,947 cm² × ~30 MPa cavity pressure implies a very large press (>3000 t) — no such machineId in the provided list; mach-haas-vf2 is a placeholder only.
4. Draft angle range unresolved (range=?°) — undercut severity may increase once actual draft measured.

Stage-1 pre-selection: auto (0%) ·